Digital Twin Anchored Generation: The Geometry of Opinion, Theory of Mind, and the Ambiguity of Sequential Questioning

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***Abstract*—Models of political opinion have typically treated attitudes either as latent properties revealed by measurement or as variables that evolve through communication and influence. Distinguishing these interpretations is difficult when an interaction simultaneously elicits a response and potentially changes the respondent. We introduce a scientific formulation of Digital Twin Anchored Generation (DTAG), in which a language interface is grounded in Large Science Models (LSMs) trained on high-dimensional survey response systems. Coupled conditional models give a partially observed synthetic respondent a full *predictive phenotype*: conditional distributions for unasked questions and a geometry determined by the consequences of responses for other attitudes. We formalize sequential question selection and its induced geometric displacement and argue that DTAG constitutes a restricted, falsifiable computational theory of mind rather than a generic language-model persona. The central dichotomy is between *phenotypic refinement*, in which a sequence reveals coordinates of a fixed respondent, and *phenotypic construction*, in which elicitation and exposure alter the subsequent state. Both can produce apparent ideological trajectories, but they have different experimental implications. Connecting this question to the evidence for backfire under cross-cutting exposure reported by Bail *et al.* (PNAS, 2018), we specify a prospective, randomized, preregisterable study suitable for Prolific, with reversed-order questioning, matched controls, held-out outcomes, and delayed remeasurement. We report completed native GSS 2024 simulations of six pinned synthetic personas: 960 adaptive ten-question trajectories and 2,880 fixed-set question-order replays, with three data-derived PGFPlots figures. Right-derived ten-item sets exhibit frequent pathwise endpoint and movement-direction reversals under reordering, whereas neutral-derived sets are invariant under the tested coupling. These model-based results do not establish human persuasion; a prospective randomized human validation remains proposed.**

***Index Terms*—Computational theory of mind, opinion dynamics, large science models, digital twins, survey methodology, social influence, polarization, latent geometry, adaptive questioning.**

Significance

Asking someone a question can reveal an existing attitude, but it can also alter what that person subsequently reports or believes. We describe a computational framework that makes this ambiguity measurable. DTAG grounds a conversational digital twin in a learned high-dimensional model of interconnected survey opinions rather than relying exclusively on plausible language generation. It predicts the responses a represented person might give to unasked questions and quantifies the geometric consequences of sequential answers. We propose a randomized human study that reverses question order, measures independent downstream attitudes, and follows respondents over time. The experiments distinguish predictive reconstruction from evidence of interaction-induced belief change.

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I. Introduction

The appropriate unit of analysis for a belief system may be neither a single attitude nor a predetermined left–right score, but the pattern of constraints among many expressed positions. Economic preferences, religious commitments, institutional trust, social identities, and political opinions can be related in ways not apparent in a short attitude battery. Classical arguments about the coherence of mass belief systems [**?**], [**?**] and subsequent measurements of partisan sorting and issue alignment [**?**], [**?**] expose the limitations of treating the public as a collection of isolated responses. The complementary problem is temporal: a person's answers may reflect a relatively stable configuration, but they may also be constructed or revised as questions, considerations, or social information are encountered.

We study the boundary between these accounts using Digital Twin Anchored Generation (DTAG). DTAG couples natural-language interaction to a Large Science Model (LSM), which represents empirically estimated conditional dependencies among survey variables. An observed answer restricts the represented phenotype and changes predictions about unasked attitudes. The model can also compute the displacement of its conditional state in a pole-relative opinion geometry and choose questions that maximize expected displacement. This permits mathematically explicit experiments with sequences of questions, rather than relying on the plausibility of generated prose.

The observation that the synthetic twin moves along an ideological coordinate does *not*, by itself, establish that a person has been persuaded. Yet the opposite claim—that questioning can only reveal fixed attitudes—is equally

unwarranted. We develop a model-level dichotomy: *passive refinement* of a pre-existing phenotype versus *reflexive construction* of the phenotype by the interaction. We ask whether learned population geometry is useful only for the former, or also constrains the latter. The distinction is empirically resolvable only by predicting outcomes of controlled human interactions before those outcomes are observed.

II. Prior Work and the Scientific Gap

A. Belief systems, cognitive organization, and survey response

Converse [**?**] emphasized weak issue constraint outside politically sophisticated subsets of the public; Zaller [**?**] offered a receive–accept–sample account in which answers depend on considerations accessible at the time of response. Later analyses distinguished partisan sorting from genuine cross-issue alignment [**?**] and showed how homophily and influence can amplify weak initial affinities across otherwise unrelated domains [**?**]. The sociology-of-knowledge perspective similarly places an individual proposition within a larger socially situated worldview. These accounts motivate measurement of the *full conditional organization* of attitudes rather than presupposing a low-dimensional ideological axis.

Cognitive psychology complicates the interpretation of measurement. Consistency pressures may promote cognitive revision [**?**]; wording, scale, question order, and conversational context change survey reports [**?**]; selective evaluation of political information can reinforce existing commitments [**?**]. Measurement reactivity does not necessarily imply enduring persuasion. A response might be an instantaneous reconstruction, temporary salience shift, response-scale artifact, or genuine revision. These alternatives predict different persistence, cross-domain generalization, and order effects and must be disaggregated experimentally.

B. Quantitative influence and experimental evidence

DeGroot's averaging process [**?**] and Friedkin–Johnsen dynamics [**?**] specify explicit interpersonal transitions; bounded-confidence accounts introduce state-dependent interaction neighborhoods [**?**]. Such models characterize population-level evolution but often simplify the *internal* state of each individual to one or a few quantities. DTAG takes a complementary route: it learns within-person conditional structure at high dimension but presently lacks an experimentally identified transition operator for real human belief change.

The discrepancy matters in the setting studied by Bail *et al.* [**?**]. Their 2018 *PNAS* field experiment randomly offered Twitter users incentives to follow bots relaying cross-cutting political content. Republicans assigned to follow a liberal bot reported more conservative views afterward; the corresponding liberalizing estimate among Democrats assigned to a conservative bot was not statistically significant. Exposure to disagreement thus did not yield a simple convergence law. This is a motivating example of potentially state-dependent and asymmetric responses, *not* evidence that DTAG identifies the causal mechanism or reproduces the reported effect. Other experiments complicate generalization: a synthesis of campaign-contact field experiments reported very small average general-election effects [**?**], whereas some structured interpersonal conversations produced more persistent changes in measured attitudes [**?**]. The need is for a framework that predicts *heterogeneity, indirect effects, and persistence* rather than asserting a universal influence direction.

C. Theory of mind and generative simulation

Theory of mind (ToM) concerns inference about another agent's unobserved beliefs, intentions, or desires and their consequences for observable behavior [**?**], [**?**]. Computational inverse-planning models infer latent states from observed actions under a generative account of agent behavior [**?**]. DTAG tackles a narrower but measurable ToM task: from a sparse response history, infer the probability distribution over a represented respondent's *unasked attitudes*. This operational definition avoids equating language fluency with access to an individual mind. Existing language-model approaches to synthetic respondents have demonstrated population or profile fidelity [**?**], [**?**]; their ability to mimic a subgroup distribution need not imply individual-level conditional calibration or validity under repeated interrogation. DTAG is designed to make the model state, prediction, and response provenance explicit.

III. Large Science Models and the Predictive Phenotype

A. Conditional dependency model

Let $X = (X_1, \dots, X_n)$ take values in $\mathcal{X} = \prod_{i=1}^{n} \mathcal{A}_i$, with each $\mathcal{A}_i$ a finite response alphabet and c denoting the survey population, wave, or supported context. An LSM is the family

$$\Phi_c = \{\phi_{i,c}(a \mid x_{-i}) : i = 1, \dots, n\}, \tag{1}$$

estimated with coupled conditional inference trees [**?**], [**?**], [**?**]. A tree dependency $j \to i$ indicates predictive relevance of X_j for X_i, not an identified causal edge. Because conditionals are estimated separately, exact compatibility with one joint distribution is not assured [**?**].

The state of a digital respondent is a partial assignment $s = \{X_j = a_j : j \in S\}$, with unknown entries represented as missing to the native runtime. For an unasked question $v \notin S$, let

$$p_\Phi(a \mid s, v) = \phi_{v,c}(a \mid \widetilde{x}_{-v}(s)). \tag{2}$$

The vector

$$\mathcal{B}_\Phi(s) = \big(p_\Phi(\cdot \mid s, v)\big)_{v \notin S} \tag{3}$$

is the *predictive phenotype*. It is more informative than a generated biography or a scalar ideology score: it records the model's distribution over as-yet-unobserved responses, including uncertainty and cross-domain conditional coupling.

B. Geometry as predictive equivalence

The dependency structure induces a geometry of possible opinions. For complete or partially represented states x, y, a representative version of the LSM distance is

$$\theta_\Phi(x,y) = \frac{1}{n}\sum_{i=1}^{n}\sqrt{D_{\mathrm{JS}}(\phi_{i,c}(\cdot \mid x_{-i}), \phi_{i,c}(\cdot \mid y_{-i}))}. \quad (4)$$

Here D_{JS} denotes Jensen–Shannon divergence; the native code evaluates the corresponding *qdistance*. Distinct states may be at distance zero when their conditional consequences are indistinguishable, so the geometry is generally pseudometric. Crucially, this is not a psychological assertion that the human mind occupies a unique Euclidean manifold: it is a testable geometry of *predictive consequence*. Differences in a few answers can change predictions across the wider response network; equally, some recorded differences have negligible conditional consequences.

The prior opinion-fabric analysis [**?**] reports held-out reconstruction from 80% masked response items. In the reported evaluation, mean LSM-distance distortion declined by approximately 54.9% (GSS) and 60.9% (Eurobarometer) relative to the masked representation. Those are geometric reconstruction results, not percentages of perfectly recovered item labels and not estimates of causal influence.

For GSS models with registered left and right reference states s_L, s_R, DTAG reports

$$I_\Phi(s) = \frac{\theta_\Phi(s_L, s) - \theta_\Phi(s_R, s)}{\theta_\Phi(s_L, s_R)}. \quad (5)$$

The reference poles establish orientation, while the conditional system establishes distance. Pole choice and wave dependence must be included in any robustness analysis; the current index should not be interpreted as a calibrated universal political ideology scale.

IV. DTAG: A Conditional Theory-of-Mind Interface

A. Architecture and generative semantics

A semantic interpreter maps natural-language question q to one or more survey variables $M(q)$. The LSM provides native response probabilities; the resulting categorical anchor is recorded, the predictive phenotype is recomputed, and a language model expresses the anchored answer in prose. For a single item,

$$A_t \sim p_\Phi(\cdot \mid s_t, v_t), \qquad v_t \in M(q_t), \quad (6)$$

$$s_{t+1} = s_t \cup \{v_t = A_t\}, \quad (7)$$

$$\mathcal{B}_{t+1} = \mathcal{B}_\Phi(s_{t+1}). \quad (8)$$

These equations describe the idealized *draw* mode without state eviction. The implementation also supports modal answers, multi-item semantic mappings, validated mappings, and direct item selection; these are distinct experimental conditions. The operational system uses a fixed LSM during a respondent session. The ordinary update is statistical conditioning, not an identified model of psychological persuasion.

B. Sequential geometry and adaptive questioning

For every candidate item v and response a, define

$$\Delta_v(a; s) = I_\Phi(s \cup \{v = a\}) - I_\Phi(s), \quad (9)$$

$$\mu_v(s) = \sum_{a \in \mathcal{A}_v} p_\Phi(a \mid s, v)\Delta_v(a; s), \quad (10)$$

$$\sigma_v^2(s) = \sum_a p_\Phi(a \mid s, v)(\Delta_v(a; s) - \mu_v(s))^2. \quad (11)$$

One-step selection can maximize or minimize $\mu_v(s)$ under stochastic answering. A separate modal-response objective instead evaluates the response with highest conditional probability. This matters: following a sequence of modes is not equivalent to sampling a population of coherent respondents. The current optimizer enumerates possible response assignments in the native LSM and evaluates their exact geometric displacement *relative to the fitted model*; it does not compute the best intervention for changing a real person's beliefs.

A companion predictive-landscape statistic measures the change in *unasked* response distributions after conditioning on one answer:

$$R_j(s; v, a) = D_{\mathrm{JS}}\big(p_\Phi(\cdot \mid s, j), p_\Phi(\cdot \mid s \cup \{v = a\}, j)\big). \quad (12)$$

A total propagation score can be defined as $\mathcal{P}_\Phi(s; v, a) = \sum_{j \notin S \cup \{v\}} w_j R_j(s; v, a)$ for prespecified nonnegative weights w_j. It measures statistical redistribution of conditional predictions, not a causal cascade of actual opinions.

C. Difference from RAG and language-model personas

Retrieval-augmented generation (RAG) supplies relevant text to a generator, primarily addressing factual grounding and provenance [**?**]. Demographically prompted or autobiographically conditioned LLM personas seek plausible behavior or distributional survey fidelity [**?**], [**?**]. DTAG asks a different question: can the model's *entire conditional distribution of unasked responses* be validated against a real respondent as anchors accumulate? The native categorical model, learned from social observations, is a state-dependent probability mechanism; the LLM is a semantic interface. RAG can complement DTAG with source context, but retrieval of passages does not, by itself, supply an empirically calibrated joint opinion model. DTAG conversely does not claim the broader language understanding or fact retrieval capabilities of RAG.

V. The Interpretation of Sequential Questions

A. Phenotypic refinement: observing a fixed system

Under the passive account, a respondent possesses a complete, fixed phenotype $X^\star$, and each response reveals information about it. A sequence $s_0 \subset s_1 \subset \cdots \subset s_T$ gives progressively more constrained predictions of unasked variables. For a designated held-out target set Y, the ideal information value of query v is

$$\mathcal{G}(v \mid s) = H(Y \mid s) - \mathbb{E}_{A_v|s}[H(Y \mid s, v = A_v)]. \quad (13)$$